$n = 2$

M=39

nearest distance 2r = 4.52

$$a = \frac{4r}{\sqrt{3}} = \frac{2 \times 4.52 \times 10^{-10}}{\sqrt{3}} = 5.21 \times 10^{-10}$$

$$\rho = \frac{2 \times 39}{\left(5.21 \times 10^{-10}\right)^3 \times \left(6.023 \times 10^{23}\right)}$$

$\rho = 915 \text{ Kg m}^{-3}$

N_A

19. b) equal number of cations and anions are missing from the lattice

20. c) Frenkel defect

21. d) Both assertion and reason are false

22. b) FeO

23. a) XY_8

UNIT-7

Choose the correct answer:

1. option (c)

$$k = \left(\frac{2.303}{t}\right) \log\left(\frac{[A_0]}{[A]}\right)$$

$$k = \left(\frac{1}{t}\right) \ln\left(\frac{[A_0]}{[A]}\right)$$

$$e^{kt} = \left(\frac{[A_0]}{[A]}\right)$$

$$[A] = [A_0] e^{-kt}$$

In this case

$k = x \text{ min}^{-1}$ and $[A_0] = 0.01\text{M} = 1 \times 10^{-2}\text{M}$

t = 1 hour = 60 min

$[A] = 1 \times 10^{-2} \left(e^{-60x}\right)$

2. option (c)

Answers.indd 242 22-12-2022 19:40:17